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ORIGINAL ARTICLE

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Upper limb cranking asymmetry during a Wingate anaerobic test in wheelchair basketball players

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Introduction: Interlimb asymmetry of strength and/or motor coordination could limit the performance of wheelchair athletes or increase their risk of injury. Studies of interlimb asymmetry in the lower limbs have shown high between-subject variability that does not depend on the side of dominance and that does not change with fatigue. Upper limb asymmetry is particularly large in manual wheelchair athletes with a lower degree of impairment. The aim of this study was to evaluate interlimb asymmetry of forces developed during an upper limb Wingate anaerobic test, the effects of fatigue on force, and differences between high- and low-point players.

Method: Twenty-five wheelchair basketball players (13 females and 12 males) of male and female national French teams performed a 30s anaerobic Wingate test on an arm ergometer. Participants were classified into two functional categories, high-point (classed from 3 to 4.5) and low-point (classed from 1 to 2.5), according to the International Wheelchair Basketball Federation classification. Left and right arm forces were measured during the pushing and pulling phases at peak power, 10s, and the end of the 30s test.

Results: Upper limb asymmetry changed with fatigue during each phase. Force asymmetry differed between peak power, 10s and 30s, with no consistent increase or decrease. Asymmetry did not differ significantly between low- and high-point players but tended to be greater in high-point players. Asymmetry tended to be greater in the females, with significant differences between the males and females in the push phase.

Conclusion: Inter-subject variability was high, but forces were asymmetric for most participants, especially females. The Wingate anaerobic test could highlight problematic asymmetries that might impact daily life or sports performance.

KEYWORDS
asymmetry, cranking, fatigue, sports, upper limbs, wheelchair, Wingate

1 | INTRODUCTION

Upper limb asymmetry is an issue for manual wheelchair users. Differences in propulsive efficiency between the dominant and nondominant sides could hinder straight-line travel and increase energy cost.[1,2] Several studies have found a large variability of interlimb asymmetry during wheelchair propulsion on a roller ergometer. They found differences in distance traveled between the right and left wheels, maximum speed reached for each wheel, and maximum power developed on the wheel rim.[2,3] Asymmetry has also been found to increase with task difficulty,[1] with increasing rolling resistance[4] and on uneven ground,[5] suggesting that asymmetry increases with the intensity of the effort exerted during propulsion.

To our knowledge, no studies have evaluated the impact of fatigue on upper limb asymmetry. Studies in sports that particularly involve the lower limbs have found different effects of fatigue on asymmetry. Two studies found that asymmetry of force developed during a unilateral countermovement jump increased with fatigue,[6,7] whereas no effect of fatigue was found on asymmetry of isometric hip extensor strength,[8] knee extensor strength,[9] thigh muscle strength,[10] or propulsive power.[11] Spatiotemporal asymmetry does not change with fatigue during running[12] or walking.[13]

In paralympic sports, players are classified according to their functional abilities to ensure fairness between competitors and in the composition of teams for sports such as wheelchair rugby[14] and wheelchair basketball (WBB).[15] In a study in 2018, Goosey-Tolfrey et al. divided a sample of wheelchair rugby players into two subgroups, low-point (LP, ranked from 0.5 to 1.5) and high-point (HP, ranked from 2.0 to 3.5).[3] They found less upper limb asymmetry in LP than HP players in terms of distance traveled and power developed on a roller ergometer, and they theorized that this could be due to the greater upper extremity demands highlighted by the higher power developed by HP players. In female wheelchair basketball, higher power developed by HP was observed in a recent study.[16] However, another study observed no significant differences in performances on bench press and a repeated modified agility test between LP and HP in wheelchair basketball.[17]

Hutzler[18] and Coutts[19] found that performance in Paralympic sports such as wheelchair basketball (WBB) or wheelchair racing depends considerably on power and maximum anaerobic capacity. Arm crank ergometers were used in studies, to assess physical fitness and performance in standardized laboratory conditions in athletes with different types of disabilities.[20–24] To our knowledge, no studies have evaluated interlimb asymmetry of the upper limbs during an arm cranking ergometer test. In able-bodied people performing the 30s anaerobic Wingate test (WAnT_30s), which is a maximum effort test, metabolism in the first 7 to 9-s is predominantly alactic and anaerobic by phosphagen degradation, and the anaerobic lactic (glycolytic) metabolism is predominant from the 10 to 30s.[25–27] These findings lead to question of whether left–right upper limb forces might differ in athletes with disabilities depending on the predominant metabolism type used during the WAnT_30s.

The first aim of this study was to assess upper limb force asymmetry during an all-out, maximal intensity 30s anaerobic Wingate test on an arm-crank ergometer. Consistent with studies of lower limb asymmetry,[28] the authors hypothesized that asymmetry would not increase during the exercise.

In parasports, athletes present different functional capacities, so as in the study by Goosey-Tolfrey et al 2018 on wheelchair rugby players,[3] the second aim was to assess differences in asymmetry and performance between LP and HP WBB players. In accordance with the results of Goosey-Tolfrey et al 2018, the authors hypothesized that both asymmetry and performance would be greater in HP than in LP players.

Therefore, to investigate whether greater upper extremity power could explain greater asymmetry, the third aim was to compare asymmetry and performance between male and female WBB players. The authors hypothesized that these variables would be greater in males than in females.

2 | METHODS

This study was approved by our local ethics committee (N°IDRCB: 2020-A02919-30) and registered on clinicaltrials.gov (NCT04748497).

2.1 | Participants

Twenty-five elite wheelchair basketball players (13 females and 12 males) from the national team took part in this study (Table 1). Excluding training national team, they trained 4 days a week with matches during weekends. Inclusion criteria were the use of a manual wheelchair for sport, and no current back or upper limb pain or injury. Participants' pathologies did not impact their upper limbs: complete ($n=14$) and incomplete ($n=4$) paraplegia, lower limb amputation ($n=3$), lower limbs congenital malformation ($n=2$), spina bifida ($n=1$), and sequelae of poliomyelitis ($n=1$). The time since disability reported in Tables 1 and 2 includes individuals with acquired and congenital disabilities. Participants were classified into two sport-specific functional categories according to the

TABLE 1 Participant characteristics.

Participants characteristics	Total ($n = 25$)	Males All ($n = 12$)	Males LP ($n = 6$)	Males HP ($n = 6$)	Females All ($n = 13$)	Females LP ($n = 7$)	Females HP ($n = 6$)
Age (years)	29 ± 7	30 ± 7	30 ± 7	29 ± 8	28 ± 7	29 ± 6	27 ± 9
BM (kg)	67 ± 11	74 ± 10	68 ± 9	80 ± 7	61 ± 7	56 ± 4	65 ± 5
Time since disability (years)	20 ± 11	23 ± 10	24 ± 11	22 ± 10	18 ± 12	12 ± 7	23 ± 13
Training age (years)	10 ± 6	12 ± 5	13 ± 5	11 ± 6	8 ± 6	6 ± 4	9 ± 8

Note: Data are mean ± standard deviation. HP = class 3.5 to 4.5; LP = class 1 to 2.5.
Abbreviations: BM, body mass; HP: high-point player; LP, low-point player.

TABLE 2 Participant characteristics according to group (low-point < 3.0 points and high-point > 2.5 points).

Participants characteristics	LP ($n = 13$)	HP ($n = 12$)	*p*-Value
Age (years)	29 ± 6	28 ± 8	0.684
Body mass (kg)	61 ± 9	73 ± 10	0.005*
Time since disability (years)	23 ± 11	18 ± 11	0.375
Training age (years)	9 ± 5	10 ± 7	0.664

Note: Data are mean ± standard deviation.
Abbreviations: HP, high-point player; LP, low-point player.
*Statistically significant values ($p < 0.05$) with an independent sample *t*-test between LP and HP.

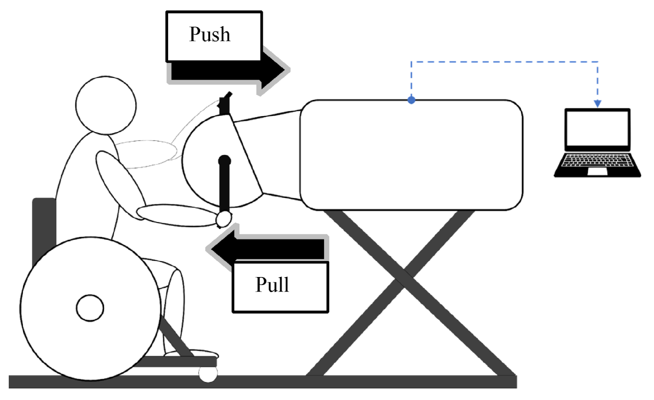

FIGURE 1 Representation of the experimental set-up on the crank ergometer.

International Wheelchair Basketball Federation classification (IWBF, 2021) The rating ranged from 1 to 4.5, with lower ratings (low-point [LP] players: classed from 1 to 2.5) indicating lower levels of sport-specific functional capacity and higher ratings (high-point [HP] players: classed from 3 to 4.5) indicating higher levels of mobility (Tables 1 and 2). All the players were evaluated by International Paralympic Committee (IPC) classifiers. The sample was dichotomized because class 3.0 players and above have complete trunk movement in the vertical and forward planes.[15] This affects performance since their trunk is not attached during the WAnT_30s on the arm crank ergometer, and the muscles that stabilize and perform trunk rotation and flexion/extension are also involved in the crank movement. The dominant side was identified subjectively by asking participants, "Which is your strongest side?"

2.2 | Materials

In this study, a commercially available arm crank ergometer (Brachumera, lode®, DEV-CA925901) was used to record the torque developed on each crank (right and left) in Newton meters (Nm) and the speed of rotation in degrees per second (°/s) or rotations per minute (RPM). This device recorded at 1000 Hz and exported values every two degrees. Resistance was generated by a Lanooy electromagnetic braking mechanism (eddy currents) and measured either in watts or by a torque factor that accounted for the participant's mass. The arm cranks were positioned asynchronously (i.e., at 180° to each other).

2.2.1 | Protocol

The protocol took place during an in-season period on a national team training camp. Participants performed an all-out, 30 second Wingate test (WAnT_30s) adapted for the upper limbs.[29] The ergometer was placed on a height-adjustable table. Participants used their basketball wheelchairs (Figure 1). Each participant was strapped to the wheelchair at their thighs or hips according to their pathologies, and the wheelchair was fixed with ratchet straps to the floor to prevent movement between the participant and the wheelchair and between the wheelchair and the floor during the test. The axis of rotation of the cranks was aligned with the glenohumeral joint.[29] The warm-up consisted of 3 min of self-selected cadence cranking at a resistance of 25 W followed by 3 min of low-intensity interval work with active rest periods (working and resting resistance of 100 and 15 W, respectively). At the end of the warm-up, participants performed a sub-maximal sprint without resistance for 15s to accustom them to the movement with a high cranking speed. The resistance value of the WAnT_30s was customized according to Cavedon et al. (2021).[21] A torque factor (torque value divided by the body mass) was a parameter in the arm ergometer software to establish resistance level for the WAnT_30s. To customize resistance for each participant, an isokinetic test was then performed at 80 rpm for 5s at peak power so that a torque factor (F-ISO) could be calculated as the maximum torque developed divided by the participant's body mass. The WAnT_30s was performed with a torque resistance factor equal to half of F-ISO and from zero velocity. The researchers provided active encouragement to participants throughout the test to ensure maximum performance.

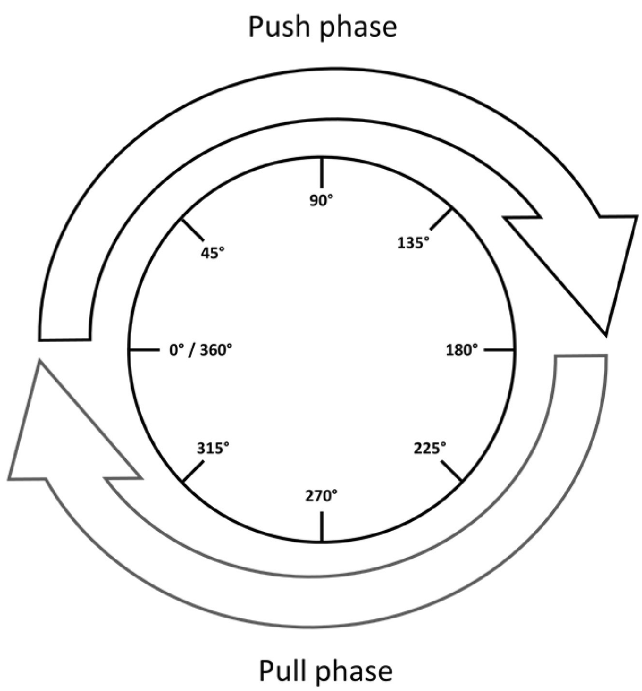

FIGURE 2 Crank Cycle Breakdown with the push phase represented by the arrow at the top and the pull phase represented by the arrow at the bottom.

2.2.2 | Data processing

The output signals obtained from the hand crank ergometer software (Lode Ergometry Manager 10th version) were processed with MATLAB software, v2020b (MathWorks Inc.). Moments of force were measured for the left (LM) and right (RM) cranks. The tangential force produced on the left (LF) and right (RF) cranks was calculated with Equation 1, where CL and CR are the lengths of the left and right cranks, respectively.

$$LF = \frac{LM}{CL} \text{ and } RF = \frac{RM}{CR} \tag{1}$$

Left (LPow) and right (RPow) power were calculated as the product of the force torque with the rotational velocity of the cranks (VR) (Equation 2).

$$LPow = CL * VR \text{ and } RPow = CR * VR \tag{2}$$

The authors also divided the crank stroke into two phases. The push phase was defined as beginning at the proximal dead center (PDC) between the end of elbow flexion and the beginning of elbow extension to the distal dead center (DDC) and ending between the end of elbow extension and the beginning of flexion. The pull phase was defined as the remaining half crank cycle (Figure 2).

The total force asymmetry (SItot) was calculated using Equation 3.1 according to Robinson et al. (1987) used on kinematic gait analysis for the comparison with other studies.[28] According to Smak et al. (1999) who compute cycling joint force asymmetries for positive and negative values separately, the authors calculated asymmetry during the pulling (SIpull) and pushing (SIpush) phases (Equations 3.2 and 3.3, respectively).[29] To compare the evolution during the WAnT_30s without the impact of interparticipant sign difference, the absolute was calculated value of the difference between the forces on each side. With these equations, a symmetry index of 0% indicates perfect symmetry.

$$SItot = \frac{|LFtot - RFtot|}{0.5 * (LFtot + RFtot)} * 100 \tag{3.1}$$

$$SIpull = \frac{|LFpull - RFpull|}{0.5 * (LFpull + RFpull)} * 100 \tag{3.2}$$

$$SIpush = \frac{|LFpush - RFpush|}{0.5 * (LFpush + RFpush)} * 100 \tag{3.3}$$

Here, LFtot and RFtot represent the average of left and right tangential forces during a full crank turn. LFpull and RFpull represent the mean force on the left and right cranks during each pulling phase and LFpush and RFpush represent the mean force on the left and right cranks during each pushing phase.

To evaluate changes in asymmetry during each phase, asymmetry was calculated at three instants of the WAnt_30s. At each instant, mean LFtot, RFtot, LFpull, RFpull, LFpush, and RFpush for 5 consecutive crank turns were calculated. The first instant was when the participant reached peak power (A_PP), which occurred during the first 5s of the test for all athletes. The second was 10s after the start of the test (A_10s), and the third was at the end of the 30s test (A_30s).

The following variables were also calculated:

- Torque factor: the personalized resistance used during the test.
- Absolute (W) and relative (W/Kg) peak and mean power, and power at 10s and 30s
- Fatigue index: percentage power loss at the end of the 30s test, as a proportion of peak power, calculated with Equation 4.

$$Fatigue\ index = \left(1 - \frac{End\ power}{Peak\ power}\right) * 100 \qquad (4)$$

- 10s fatigue index: the same as the fatigue index but calculated between peak power and 10s after the beginning of the test. In this case, in Equation 4, End power represents 10s power.
- Rotational velocity of the cranks at peak power, at 10s and at the end of the test, in revolutions per minute.

Peak power and velocity were obtained during the first 5s of the test, and end power and velocity were obtained during the last cycle of the 30s test.

2.3 | Statistical analysis

Numerical variables are presented as mean ± standard deviation, and the normal distribution of the data was tested with the Shapiro–Wilk test. The homoscedasticity of the data was calculated with Levene's test. Nonparametric Wilcoxon test or *t*-test were used for paired samples accordingly. The Mann–Whitney U-test or an independent sample *t*-test was used to compare results between the LP and HP groups and the men and women groups. The authors used the paired *t*-test or Wilcoxon signed-rank test to compare asymmetry magnitude between the three instants of the test, and variation of asymmetry and performance between LP and HP groups and males and females. The authors used the 26th version of Statistical Package for the Social Sciences (SPSS 26) software for all analyses and considered $p < 0.05$ as statistically significant.

3 | RESULTS

By definition, the functional capacity of LP and HP players differs (IWBF, 2021), but there are also differences in mass where HP players were heavier ($p = 0.005$). Age was similar in the groups (Tables 1 and 2). Three players declared themselves stronger on the left side and 22 declared themselves stronger on the right side. The peak power was obtained at 4.2 ± 1.7 s after the start of test on average.

Table 3 shows the comparison of absolute performance between male and female participants. Torque factor (i.e., the resistance provided during the test) differed significantly between males and females ($p < 0.0001$). Peak power ($p < 0.0001$ for absolute and relative power), mean power ($p < 0.0001$ for absolute and relative power), and power at 10s ($p < 0.0001$ for absolute and relative power) were higher in men than women but there was no difference for power at 30s (end of test) ($p = 0.125$ in absolute power and, $p = 0.735$ in relative power). Average force developed ($p < 0.0001$) was higher in males but there was no difference in velocity at peak power ($p = 0.203$) or at 10s time instant ($p = 0.282$). Crank speed was higher in females at 30s ($p = 0.001$). Tatigue index was higher in males ($p = 0.002$).

Comparison of HP and LP groups found no significant difference in mean force (HP: 231.1 ± 100.8 and LP: 188.7 ± 64.67, $p = 0.270$), mean power (HP: 284.3 ± 93.5 and LP: 249.2 ± 81.9, $p = 0.247$), or peak power (HP: 472.2 ± 178.35 and LP: 370.6 ± 141.14, $p = 0.650$). Relative power at 30s was higher in the LP group (HP: 2.08 ± 0.87 and LP: 2.75 ± 0.44, $p = 0.022$).

Within the male and female subgroups, variables did not differ between the LP and HP groups except for the fatigue index at 10s in females ($p = 0.031$), which was higher for the HP group. For all participants, mean power was significantly higher during the first 10s than throughout the 30s test ($p < 0.0001$).

Regarding the percent of total force asymmetry at the three time-instants considered, differences were found between peak power and 10s (Table 4). Mean asymmetry increased for the whole sample ($p = 0.028$) and for the HP group ($p = 0.022$). Force asymmetry tended to increase in the whole sample and in the LP and HP groups, despite considerable between-participant variability. Although not significant, asymmetry decreased from the 10s to the 30s measurements. Pushing and pulling force asymmetry did not change significantly, although they were not constant during the test.

Asymmetry tended to be higher in the HP than in the LP group, although the difference was not significant (Table 5).

TABLE 3 Comparison of variables between men and women and high-point (HP) and low-point players (LP).

			Men			Women		
		Total (n = 25)	**All (n = 12)**	**LP (n = 6)**	**HP (n = 6)**	**All (n = 13)**	**LP (n = 7)**	**HP (n = 6)**
Torque factor		0.52 ± 0.23	**0.69 ± 0.22**	0.62 ± 0.1	0.8 ± 0.29	**0.36 ± 0.08***	0.38 ± 0.08	0.34 ± 0.09
Peak power	Absolute (W)	419.4 ± 165	**567.6 ± 101.6**	505.0 ± 81.5	630.2 ± 81.7	**282.5 ± 52.3***	255.3 ± 26.9	314.3 ± 58.7
	Relative ($W.Kg^{-1}$)	6.2 ± 1.8	**7.7 ± 1.2**	7.4 ± 0.7	8.1 ± 1.6	**4.7 ± 0.8***	4.6 ± 0.6	4.8 ± 1
Mean power	Absolute (W)	266 ± 87.6	**345.6 ± 53.2**	327.1 ± 48.6	364.1 ± 55.1	**192.6 ± 24.6***	182.5 ± 13.9	204.4 ± 30.2
	Relative ($W.Kg^{-1}$)	3.9 ± 0.9	**4.7 ± 0.6**	4.8 ± 0.5	4.6 ± 0.6	**3.2 ± 0.4***	3.3 ± 0.4	3.1 ± 0.5
10s Mean power	Absolute (W)	309.7 ± 108.4	**409.7 ± 60.6**	404.0 ± 52.2	415.5 ± 72.6	**217.4 ± 30.1***	204.1 ± 17.0	232.9 ± 36.0
	Relative ($W.Kg^{-1}$)	4.6 ± 1.1	**5.6 ± 0.6**	5.4 ± 0.5	5.7 ± 0.7	**3.6 ± 0.5***	3.7 ± 0.4	3.6 ± 0.6
End power	Absolute (W)	159.9 ± 56.7	179.3 ± 76.1	207.1 ± 27.6	151.5 ± 100.6	142.0 ± 19.8	137.7 ± 11.3	147.1 ± 27
	Relative ($W.Kg^{-1}$)	2.4 ± 0.7	2.5 ± 1	3.1 ± 0.4	1.9 ± 1.2	2.4 ± 0.3	2.5 ± 0.3	2.3 ± 0.4
10s Power	Absolute (W)	261.5 ± 98.5	**343.3 ± 77.4**	322.6 ± 81.5	364.0 ± 74.2	**186.0 ± 32.5***	183.1 ± 20.8	189.4 ± 44.6
	Relative ($W.Kg^{-1}$)	3.9 ± 1.1	**4.7 ± 0.9**	4.4 ± 1.0	5.0 ± 0.8	**3.1 ± 0.6***	3.3 ± 0.5	2.9 ± 0.7
Fatigue index	%	57.4 ± 14.7	**66.7 ± 15.1**	58.8 ± 2.5	74.5 ± 18.6	**48.8 ± 7.8***	45.7 ± 6.1	52.5 ± 8.5
10s Fatigue index	%	35.7 ± 12.3	38.2 ± 14.5	43.9 ± 16.9	32.4 ± 9.8	33.4 ± 9.9	**28.1 ± 6.6**	**39.5 ± 10.0**[a]
Peak velocity	RPM	87.2 ± 13.9	91.1 ± 18.3	93.1 ± 8.7	89.1 ± 25.5	83.6 ± 7	80.5 ± 7.8	87.1 ± 4.1
10s Velocity	RPM	85.2 ± 22.4	80.1 ± 29.2	71.0 ± 33.9	89.2 ± 23.1	89.9 ± 12.9	87.9 ± 11.7	92.3 ± 14.9
End velocity	RPM	71.5 ± 18.4	**59.9 ± 13**	58.7 ± 7.4	61.1 ± 17.7	**82.2 ± 16.2***	80.2 ± 15.7	84.4 ± 18
Mean force	*n*	209.1 ± 84.9	**280.8 ± 65.8**	247.9 ± 41.5	313.6 ± 72.1	**142.9 ± 24.1***	138 ± 20.7	148.6 ± 28.4
Mean push force	*n*	94.7 ± 34.4	**120.6 ± 28.4**	109.0 ± 18.9	132.3 ± 32.9	**70.7 ± 18.1***	62.5 ± 13.9	80.2 ± 18.6
Mean pull force	*n*	113.8 ± 56.1	**159.7 ± 43.5**	137.6 ± 26	181.7 ± 48.1	**71.4 ± 22.6***	75.0 ± 22.2	67.3 ± 24.3

Note: Data are mean ± standard deviation. Values in bold are data between which there are significant differences.

Abbreviations: HP, high-point player; LP, low-point player.

*Statistically significant values ($p < 0.05$) with an independent sample t-test between men and women ($p < 0.05$).

[a]Statistically significant values ($p < 0.05$) between LP women and HP women ($p < 0.05$).

TABLE 4 Comparison between symmetry index (SI %) of total force, push force and pull force developed at peak power (A_PP), 10s (A_10s), and the end of the test (A_30s) for the whole sample and LP and HP wheelchair basketball players.

	A_PP	A_10s	A_30s	*p*-Value		
SI (%)	M ± SD	M ± SD	M ± SD	A_PP vs. A_10s	A_PP vs. A_30s	A_10s vs. A_30s
Total force ($n = 25$)	12.5 ± 10.5	17.1 ± 14.8	14.5 ± 12.1	0.028[a,*]	0.313	0.696
Push force	14.8 ± 12.8	15.6 ± 11.9	17 ± 12.1	0.581	0.339	0.563
Pull force	12.9 ± 11.8	17.8 ± 12.5	16.9 ± 14.1	0.058	0.104	0.925
Total force LP ($n = 13$)	11.6 ± 11	15.2 ± 14.3	11.3 ± 10.8	0.263	0.6	0.807
Push force LP	16.8 ± 14.7	13.6 ± 10.2	14.2 ± 9.7	0.463	0.972	0.879
Pull force LP	11 ± 9.4	17.3 ± 10.3	14.6 ± 12.4	0.101	0.173	0.587
Total force HP ($n = 12$)	13.5 ± 10.3	19.1 ± 15.7	18 ± 13	0.022[b,*]	0.259	0.79
Push force HP	12.6 ± 10.6	17.7 ± 13.6	20.1 ± 14	0.099	0.132	0.638
Pull force HP	15 ± 14.1	18.3 ± 15.1	19.3 ± 15.8	0.308	0.347	0.861

Note: Data are mean ± standard deviation.

Abbreviations: HP, high-point player; LP, low-point player; A_PP, at the peak power instant; A_10s, 10s after the beginning of the test; A_30s, the end of the test; SI: Symmetry Index.

*Statistically significant values ($p < 0.05$) with [a]Mann–Whitney U-test or [b]Independent sample *t*-test between A_PP A_10s or A_30s.

TABLE 5 Comparison of mean symmetry index (SI %) all along the WanT_30s of total force, push force and pull force between high points players (HP) and low points players (LP).

	HP	LP	
SI (%)	M ± SD	M ± SD	*p*-Value
Total force	13.7 ± 11	10.8 ± 7.9	0.078
Push force	14.3 ± 10.8	13.5 ± 9.3	0.225
Pull force	14.0 ± 12.3	12.5 ± 11.6	0.060

Abbreviations: HP, high-point player; LP, low-point player; M, mean, SD, standard deviation, SI: Symmetry Index.

Within the male and female subgroups, total force asymmetry also tended to increase between A_PP and A_10s and to decrease between A_10s and A_30s. Pull force asymmetry increased significantly between A_PP and A_10s in the females (Table 6). Total force asymmetry (respectively for males and females, 10 ± 8.1, 14.1 ± 10.5, $p = 0.376$) and pull force (respectively for males and females, 13 ± 12.5, 13.4 ± 11.5, $p = 0.689$) tended to be higher in females than males, and this difference was significant for push force asymmetry (respectively for males and females, 9 ± 7, 18.5 ± 10.1, $p = 0.014$).

Over the entire test, nine participants developed more force on the left and 16 developed more force on the right. Visual analysis of the data at each instant studied and for each phase of the crank cycle found a reversal of the dominant side for total force during the 30s of the test for five participants, during the push phase for eight and during the pull phase for 12 (Figure 3).

4 | DISCUSSION

This study compared the evolution of symmetry index of force (total, push and pull) between the right and left sides during a 30s anaerobic Wingate test on a crank ergometer in national male and female wheelchair basketball players. This study also compared differences in performance and asymmetry between level of classification (HP and LP) and between men and women.

Our main results show significantly higher total force asymmetry at 10s after the start of the test than at the end. This finding shows that upper limb asymmetry fluctuates more during the first 10s than at the end of the test, despite the high intra- and interparticipant variability of asymmetry. The greatest asymmetry was found at A_10s and could show a link between the onset of fatigue and the peak of muscle imbalance. This leads us to believe that asymmetry is not related to anaerobic capacity but is more likely to the player's motor coordination and trunk control. HP players tended to be more asymmetric than LP players, although the difference was not significant. Asymmetry was greater in the females than in the males, and this was significant for push force.

4.1 | Change in symmetry over time

A reversal of the symmetry index sign for total force occurred during the 30s of the test for 5 participants, during the push phase for 8 and during the pull phase for 12 (Figure 3). Similar results were found in a study of lower

TABLE 6 Comparison of the symmetry index (SI %) for total force, push force and pull force developed between peak power (A_PP), 10s (A_10s), and the end of the test (A_30s) for male and female wheelchair basketball players.

SI (%)	A_PP	A_10s	A_30s	*p*-Value A_PP vs. A_10s	A_PP vs. A_30s	A_10s vs. A_30s
Total Force Women ($n=13$)	13.4 ± 11.7	19.9 ± 16.7	16.1 ± 13.6	0.065	0.507	0.449
Push Force Women	18.8 ± 15	18.5 ± 12.5	19.5 ± 13.5	0.972	0.807	0.701
Pull Force Women	12.1 ± 11.8	22.2 ± 11.3	19.3 ± 14.7	0.004*	0.06	0.547
Total Force Men ($n=12$)	11.5 ± 9.4	14 ± 12.3	12.8 ± 12.7	0.136	0.53	0.743
Push Force Men	10.5 ± 8.6	12.4 ± 10.8	14.4 ± 10.2	0.388	0.209	0.638
Pull Force Men	13.8 ± 12.3	13 ± 12.4	14.2 ± 13.5	0.855	0.638	0.638

Note: Data are mean ± standard deviation.

Abbreviation: A_PP, at the peak power instant; A_10s, 10s after the beginning of the test; A_30s, the end of the test; SI: Symmetry Index.

*Statistically significant values ($p < 0.05$) with an independent sample *t*-test between A_PP A_10s or A_30s.

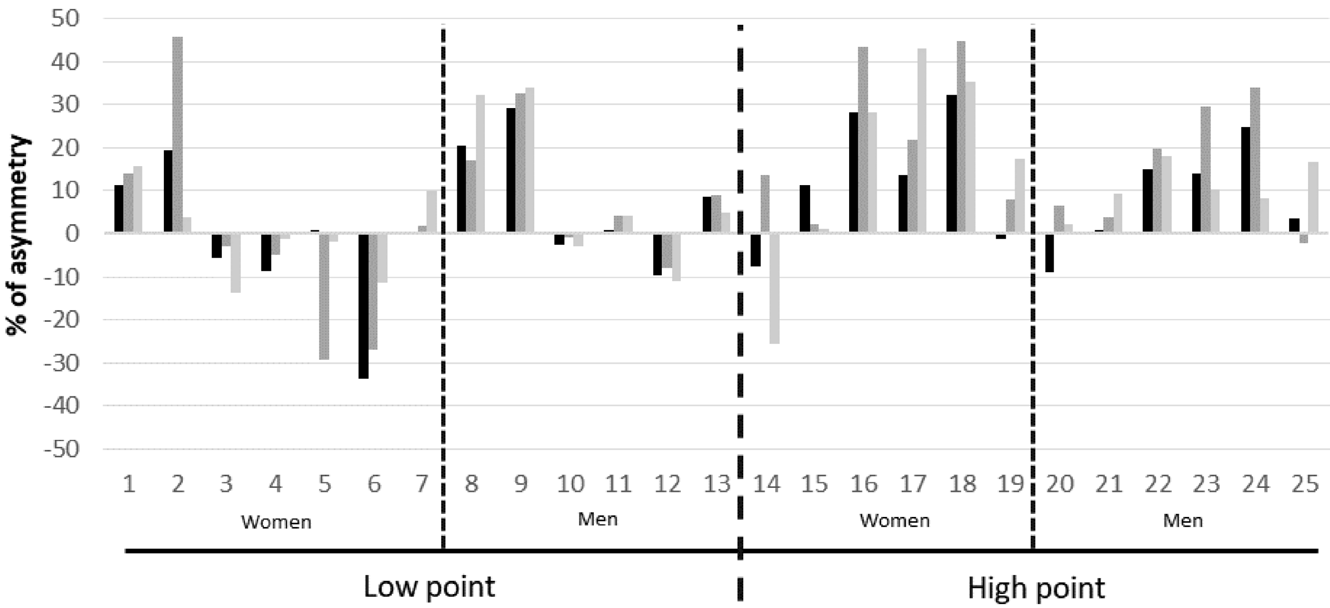

FIGURE 3 Symmetry index (%) for each participant for each time-instant evaluated: A_PP (at the peak power instant), A_10s (10s after the beginning of the test) and A_30s (the end of the test). Positive values represent right-sided dominance and negative values represent left-sided dominance. Numbers 1–25 represent each player.

limb asymmetry on a cycle ergometer[30]: the side that developed the highest force change on different test days for many participants. The authors found high inter-subject and intra-subject variability in the whole sample, but also within the LP and HP subgroups. This variability is consistent with the literature on lower extremity asymmetry during pedaling[31,32] and upper extremity asymmetry during wheelchair propulsion.[2,3]

4.2 | Upper limb force asymmetry and lateralization

The players' dominant sides were self-reported and did not correspond to the side that developed the highest forces on the machine. All three participants who reported being left-handed developed more mean force on the right. Of the 22 right-handed basketball players, seven produced more force on the left than the right during the test. Therefore, the participant's self-declared dominant side does not necessarily develop the most force during a cranking exercise at high anaerobic intensity in WBB players. Similar results have been reported in healthy people.[33,34] However, no study discussed the comparison of the self-declared dominant side with the asymmetry in wheelchair users or para-athletes during wheelchair propulsion. It would be interesting to analyze the relationship between the self-declared lateralization with the strongest or most powerful side for wheelchair propulsion or maneuverability.

4.3 | Magnitude of asymmetry

The mean asymmetry index was above 10% (13.7 ± 11% for HP and 10.8 ± 7.9% for LP), which is the threshold suggested by Carpes et al in 2007.[35] This asymmetry was larger than that found for both able-bodied and disabled athletes on an isokinetic ergometer. A recent study evaluated internal and external shoulder rotation on an isokinetic ergometer in WBB players. Although asymmetry was not specifically evaluated, an inspection of their results showed less asymmetry on the single-joint movements of internal (HP: 6%, LP: 6%) and external rotation (HP: 3.2%, LP: 8.9%) than in our study that involved multi-joint movements.[36] In a sample of able-bodied, elite volleyball players, the authors found a 7.5% asymmetry for internal rotation and 4.6% for external rotation with our equation used on data from Markou et al.[37] Asymmetry was also greater than that found in people with paraplegia or para-athletes on a roller ergometer. Calculation of an average SI with our equation using data from the study by Soltau et al. found 3.4% asymmetry of tangential force between the right and left wheels in 80 paraplegic participants during a high-speed movement.[1] In the study by Goosey-Tolfrey et al, peak power asymmetry was around 9% ± 6.7% for HP and 9.2% ± 4.4% for LP in wheelchair rugby players.[3] However, the asymmetry found in the present study was similar to that of able-bodied swimmers on a swimming ergometer (Weba) maximum force asymmetry was around 9% ± 2% for men and 8% ± 3% for women in breaststroke, and front crawl, 12% ± 5% for men and 14% ± 4% for women.[38] Another study of front crawl whose measuring forces in the swimming pool with a Globus ergometer (Globus) found an asymmetry of 19% ± 14% with a similar equation to the one the authors used.[39] Due to the complex environment, the previous result could have a higher margin of error. However, the similarity between our results and those from studies of swimmers can be explained by the multi-joint movement involving trunk rotation in our protocol too, suggesting that the asymmetry found in the present study was mainly caused by rotational asymmetry of the trunk. It would be relevant to measure the trunk rotation asymmetries in WBB players because trunk rotation is very important for maneuvering the wheelchair during matches.

4.4 | Comparison of asymmetry between HP and LP players

Asymmetry tended to be higher in the HP than in the LP group, although the differences were not significant. The authors found only two studies that analyzed differences in left–right upper limb asymmetry in disabled athletes, both in wheelchair rugby players. Neither study analyzed force asymmetry or used an arm cranking test. Goosey-Tolfrey et al. found significantly greater asymmetry in HP than in LP players for maximum speed reached between the right and left wheel on a roller ergometer but not for power.[3] Bakatchina et al. found no significant difference in speed asymmetry between HP and LP players during 20-meter sprints on a wooden floor.[40] The main differences between those studies and the present study were the populations studied and the testing mechanism (wheelchair propulsion and cranking ergometer); however, despite differences in the classification process of wheelchair rugby and WBB players, in both sports higher point classifications are aligned with higher functional capacity and HP players produce greater power and velocity than LP players.[36] Therefore, analysis by player classification in both sports allows asymmetry to be studied in two populations with more or less efficient displacement capacities in a manual wheelchair. In their studies, Goosey-Tolfrey et al and Bakatchina et al. used an asymmetry equation that compared the left/right difference on the better-performing side of the two. This is consistent because the right and left wheels are independent; thus, the individual can develop different speeds and forces on each side. The authors chose to compare the right/left difference on the average of the two sides[41] because, in our protocol, the cranks were linked. Therefore, speed was identical on both sides and only force and power varied between right and left. The main criterion for distinguishing these groups in our study was the ability to use the abdominal muscles.[15] The authors could hypothesize that trunk muscle capacity had a major impact on the performance of this test, both on the maximum force developed and the fatigue index, and that this could explain differences in asymmetry between the two studies. The authors were unable to find studies similar to our study with which to compare our results. However, it seems that differences in classification alone do not explain the tendency for higher force asymmetry in HP players, as evidenced by the nonsignificant result between HP and LP groups even though HP players had more trunk function and higher performance results.

4.5 | Comparison of power between LP and HP, and male and female players

There are no differences in power (mean power $p = 0.247$, peak power $p = 0.650$) or force ($p = 0.270$) between LP and HP players. It could be hypothesized that this homogeneity of performance was related to the lack of an increase or decrease in asymmetry across time (Table 4). The authors analyzed this separately for males and females because the males developed significantly higher forces than the females, and they had a significantly higher fatigue index (Table 3). A similar, but nonsignificant, trend between LP and HP players was found within the male and female subgroups with a higher absolute peak power and fatigue index for HP players. Despite this trend, performance of the LP and HP groups was similar and was distinguished by a higher body mass for the HP players (Tables 1 and 2). The performances of the female group was similar to that found by Molik et al.[42] and the performances of the male group were similar to the male team in the Molik et al 2010 study.[43] In 2013, these authors found significant differences between LP and HP players for peak and average power, which the authors did not find in the female subgroup in the present study. There were less differences between the LP and HP groups and these differences were not significant in the present study. Cycle ergometer resistance was determined by body weight in those studies, whereas the authors determined the load according to the power developed by the participant during a test performed before the WAnT_30s.[21] These methodological differences could explain the difference in results between LP and HP for anaerobic performance: In addition to differences in muscle or fat mass, body weight is impacted by the absence of lower limbs[44] and muscle atrophy. In comparison with the study by Cavedon et al., our peak and average power results were greater for the males.[21]

4.6 | Asymmetry in the different phases

A crank ergometer was used to record force torque data every two degrees for each crank. This allowed us to accurately measure changes in the force developed and to analyze the different phases of the crank movement (push and pull). The authors believe that this distinction of the crank cycle phases is relevant because they involve different muscle groups. Asymmetry did not differ for the push and pull phases. The authors could assume that a person with a greater push force on one side and a greater pull force on the other would have a strength asymmetry in the rotator muscles of the trunk. Similarly, the authors could assume that a person with greater push and pull forces on the same side would have a strength asymmetry of the shoulder muscles. It would be interesting to study asymmetry during single joint movements of the upper limb and the trunk. This could identify whether there is a link between the predominant interlimb force asymmetry of a joint and the resulting asymmetry of a multi-joint movement or whether this mono-joint asymmetry was compensated by the other joints. The authors were unable to find any studies that addressed this question.

4.7 | Association between asymmetry and performance variables

A separate analysis of asymmetry for males and females showed that despite many significant differences in performance variables between subgroups, and the larger effect of fatigue on males (Table 3) asymmetry did not change in either group with fatigue (Table 6). Thus, upper limb asymmetry does not seem to be related to the level of force, power, or speed developed. A study of swimmers also found no relationship between swimming speed with a higher frequency of arm movements and coordination asymmetry.[45] In the present study, mean arm cranking velocity recorded at the beginning of the test did not differ between LP and HP players. Velocity seemed to increase to a similar extent across participants during the first 10s. There is also a greater asymmetry in females than in males with a significant difference in pull force. These results are different from those observed in the literature in healthy populations,[38,46] and in volleyball players for an isokinetic test of glenohumeral muscle strength performed in sitting.[47] This can be explained by the heterogeneity of the pathologies of the lower limbs and the trunk which thus seem to have an impact on the asymmetry of force developed by the upper body. Moreover, in our protocol, the upper limbs were not evaluated separately.

In summary, the asymmetry differed most between max power (4.2 ± 1.7 s after the start of the test) and 10s after the start of the test. The differences were significant for the total force of the total sample, for the total force of the HP players and the pull force of the females. Therefore, asymmetry appears to develop when the player can no longer increase power. It would be interesting to carry out additional analyzes to understand the mechanisms that can impact this asymmetry in the seconds following achievement of peak power in WBB players.

In our sample, few participants developed symmetrical forces between the right and left sides. It was common to observe a significant force asymmetry at one instant of the test, but care must be taken when the asymmetry is significant throughout the test, as this may indicate overloading of one side compared with the other and thus an increased risk of musculoskeletal disorders. Identification of such asymmetry would require additional evaluations with on-board technologies, such as instrumented wheels or inertial measurement units to determine whether the origin of the asymmetry is functional or technical.

4.8 | Limitations

An important limitation of this study is the relatively small population and subgroup sizes, as well as the impairment and morphological heterogeneity of para-athlete populations. These factors may limit the generalizability of our findings to other populations and may have reduced the statistical power of our analyses. In the method, the tests were performed on a cranked cycle ergometer to allow specific variables to be evaluated with no external disturbances. Thus, the comparison of different athletes could be done in terms of asymmetries and intrinsic performances, without considering wheelchair propulsion technique. This will also enable future comparison with healthy people. However, the movements studied in this study for WBB players are far from the reality of the field and wheelchair propulsion. A roller ergometer would have allowed analysis of fatigue impact on asymmetry with the gestures and the technicality of the propulsion in a manual wheelchair whose studied population would be familiar.

To further the understanding of changes in upper limb asymmetry with fatigue, changes in range of motion of the upper limbs and of the trunk could be measured. Moreover, some players moved closer to the crank ergometer during the test; thus, it would be interesting to measure trunk movement in future studies.

The analysis of the change in asymmetry with fatigue was based on the fact that, according to the literature, the WAnT_30s targets anaerobic capacity with a lactic metabolism transition around the 10th second.[25–27] However, without lactate measurement, the authors cannot be sure of the metabolism used by the participants.

4.9 | Perspectives

In summary, this study suggests that upper limb force asymmetry in WBB players changes during the Wingate anaerobic 30s arm test, with no consistent increase or decrease. Asymmetry was above 10%, although variability was very high. Player classification did not impact on the results, although HP players and females tended to have larger asymmetry than LP players and males, respectively. Future research should investigate whether the asymmetry found in those with the largest asymmetry impacts on their daily life or their sports performance: If the cause could be identified, solutions may be found. Further studies should involve anaerobic tests in ecological conditions, with quantification of trunk rotation asymmetries in addition to the upper limb asymmetries.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

INFORMED CONSENT STATEMENT

Informed consent was obtained from all subjects involved in the study.

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